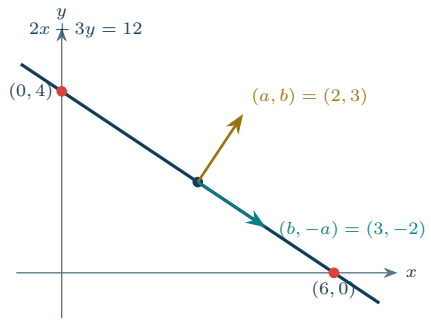
ALGEBRAIC GEOMETRY ON ONE PAGE

The line $ax + by = c$

The set of all points whose coordinates satisfy one linear equation is a straight line, and its coefficients carry the geometry.

1 / THE RULE AND THE PICTURE

An equation $ax + by = c$ picks out exactly the points (x, y) that satisfy it, and they form a line



Take $2x+3y = 12$. A point such as $(3, 2)$ lies on the line because $2 \cdot 3+3 \cdot 2 = 12$, while $(3, 3)$ does not because it gives 15. The numbers a, b and c then read off the geometry:

$$ax + by = c: \quad x\text{-intercept } \tfrac{c}{a}, \quad y\text{-intercept } \tfrac{c}{b}, \quad \text{gradient } -\tfrac{a}{b}$$

The vector (a, b) is perpendicular to the line and $(b, -a)$ points along it, so $2x + 3y = 12$ has intercepts 6 and 4, gradient $-\frac{2}{3}$ and normal $(2, 3)$.

2 / SOLUTIONS / ONE EQUATION, INFINITELY MANY POINTS

With one equation in two unknowns, one variable is free and the other follows

x	-3	0	3	6
$y = \dfrac{12 - 2x}{3}$	6	4	2	0
check $2x + 3y$	12	12	12	12

Writing $x = t$ gives the whole line as $(t, \ \frac{12-2t}{3})$, and rearranging to $y = -\frac{2}{3}x + 4$ is the familiar gradient form with $m = -a/b$ and $k = c/b$. In the language of algebraic geometry, the line is the zero set of the degree one polynomial $ax + by - c$.

3 / SCALING AND SPECIAL CASES / THE SAME LINE IN MANY DRESSES

Multiplying all three numbers by a constant changes the equation and leaves the line unchanged

$4x + 6y = 24$ is $2x + 3y = 12$ doubled, and $-2x - 3y = -12$ is the same line multiplied by -1, so the simplest form divides by the GCD of the coefficients. This is why elimination is allowed to scale an equation before combining it with another.

$c = 0$	line through the origin
$a = 0$	$by = c$, a horizontal line $y = \frac{c}{b}$
$b = 0$	$ax = c$, a vertical line $x = \frac{c}{a}$
$a = b = 0$	no line: all or no points

4 / TWO LINES / CROSSING, PARALLEL OR IDENTICAL

Two equations $ax + by = c$ and $dx + ey = f$ give a crossing, parallel lines or one repeated line



one point | no point | every point

The normals (a, b) and (d, e) are parallel exactly when $ae - bd = 0$. If $ae - bd \neq 0$ the lines cross once, at $x = \dfrac{ce - bf}{ae - bd}$, $y = \dfrac{af - cd}{ae - bd}$. If $ae - bd = 0$ the lines are parallel with no common point, or the same line when the constants agree too, as in $2x + 3y = 12$ and $4x + 6y = 24$.

5 / BUILDING THE EQUATION / THREE ROUTES TO THE SAME LINE

A point with a normal, two intercepts or two points each produce $2x + 3y = 12$

given	working	result
point $(3, 2)$ and normal $(2, 3)$	$2(x - 3) + 3(y - 2) = 0$	$2x + 3y = 12$
intercepts 6 and 4	$\frac{x}{6} + \frac{y}{4} = 1$, times 12	$2x + 3y = 12$
points $(6, 0)$ and $(0, 4)$	$6a = c$ and $4b = c$, so $a : b : c = 2 : 3 : 12$	$2x + 3y = 12$

The first route uses $c = ax_0 + by_0$ for any point (x_0, y_0) on the line, which is also the quickest check that a point lies on it.

6 / THE NORMAL VECTOR / DISTANCE FROM THE ORIGIN

The coefficients also give the distance from the origin to the line

The foot of the perpendicular from the origin lies along (a, b), and the distance is $\dfrac{|c|}{\sqrt{a^2 + b^2}}$; for $2x + 3y = 12$ it is $\dfrac{12}{\sqrt{13}} = 3.33$. A line $y = mx + k$ is at distance $|k|/\sqrt{1 + m^2}$, and a circle of radius r about the origin is touched by the line exactly when this distance equals r.

ASIDE / WHY GEOMETRY

Algebraic geometry begins here, with a polynomial equation in two unknowns defining a curve, and degree one gives the line.

ASIDE / CHECK IN PYTHON

`12/sqrt(13)` gives 3.328, and `2*3+3*2` gives 12.

Convention. Lines are written $ax + by = c$ with $(a, b) \neq (0, 0)$; the gradient is $-a/b$ when $b \neq 0$.

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